

Representation of an Agrivoltaic Farm in a Climate Model



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April 25, 2025

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Introduction

Agrivoltaics (agriPV) is a mode of agricultural and energy production in which solar panels are installed above fields. Although its practical implementation is still largely empirical, this method appears to have potential for future development in certain geographical regions. The literature on agriPV shows that crop yields can be positively affected by the implementation of agriPV systems (Rapella et al. [1], Barron-Gafford et al. [2], Al Mamun et al. [3], and Amaducci et al. [4]), while such systems could also produce a substantial amount of clean energy. Among other benefits, agriPV addresses land-use conflicts. Today, solar farms compete with agriculture for the use of arable land. An agriPV field resolves this conflict by combining agricultural production with renewable-energy generation. The question is therefore whether combining the two activities can provide satisfactory electricity and crop yields. Finally, in arid regions, the presence of solar panels in fields appears to improve water retention, at a time when water is becoming increasingly scarce in the context of global warming.

In-depth studies of agriPV systems are therefore required in order to better anticipate their potential benefits and encourage the development of this new production method.

Current studies of agriPV show that the presence of panels locally affects the climate–water–energy–food nexus. In particular, as mentioned above, they report encouraging results regarding yields, water retention, and soil depletion. Another current challenge for agriculture, however, is its need for nitrogen inputs. These inputs, which often consist of ammonium and nitrates, cause substantial soil and water pollution. The literature discusses this issue extensively [5], and it is now the subject of socioeconomic debate. Excess amounts of these products are carried by rainwater and runoff. They accumulate in lakes and reservoirs and can lead to eutrophication. They are also responsible for the emission of substantial quantities of nitrous oxide (N_2O) into the atmosphere. According to the IPCC’s AR6 report, this greenhouse gas has a global warming potential 273 times greater than that of CO_2 . Reducing the quantities of nitrogen inputs used in agriculture is therefore a major concern.

We consequently asked whether agriPV could provide a solution to this problem. Some studies [6] show that plants absorb nitrogen inputs more effectively when soil moisture is at an optimal level. In the most arid regions, agriPV systems appear to maintain higher soil and air moisture. AgriPV fields might therefore require fewer nitrogen inputs to maintain their yields.

It is thus appropriate to investigate whether the presence of solar panels in an arid region such as the Iberian Peninsula can reduce the nitrogen inputs required to maintain high yields. It is also relevant to examine nitrous-oxide emissions and the quality of plant water use in the same region. We therefore simulated agriPV systems on the Iberian Peninsula in order to study plant development as the nitrogen-input rate varies relative to a reference rate.

1 Data and methods

We use a model that simulates the effects of solar panels on the plants growing underneath them; hereafter, we refer to it as the agriPV model. The model combines a specific representation of photovoltaic panels with local-climate modelling in order to simulate plant growth from the surrounding climatic conditions.

1.1 AgriPV model

To study the agricultural production of agriPV systems, we used ORCHIDEE. ORCHIDEE is a dynamic vegetation model designed to simulate complex interactions between climate and ecosystems. Its inputs include climate data such as wind speed 1.2 m above the ground, solar radiation, and atmospheric pressure. Its outputs are simulated data describing vegetation life-cycle parameters. These outputs provide information on plant health and development, as well as on the efficiency with which resources such as water and nitrogen inputs are used.

1.1.1 PV model

Because photovoltaic panels locally alter certain atmospheric variables, a specific representation of their operation must be incorporated into ORCHIDEE. The photovoltaic (PV) module used here is based on previous work [1], itself inspired by urban and vegetation canopy models (Macdonald et al. [8]). Its main assumption is that rows of solar panels can be treated as a canopy of regularly spaced blocks, analogous to a city or a dense forest. This makes it possible to recalculate atmospheric variables within the agriPV structure. The choice is supported by experimental observations and aerodynamic analyses from fluid mechanics in porous media, in which panels are treated as periodic obstacles that influence wind-speed and temperature profiles.

Simulating the solar panels therefore involves modifying the climatic parameters locally experienced by the plants. For the model to remain as close as possible to reality, however, the effect of the panels on the different parameters must be represented realistically. The PV model mainly affects two atmospheric variables: solar radiation incident on the ground and local wind speed. These variables are then fed into the ORCHIDEE climate model together with the other atmospheric variables, as shown in the following diagram.

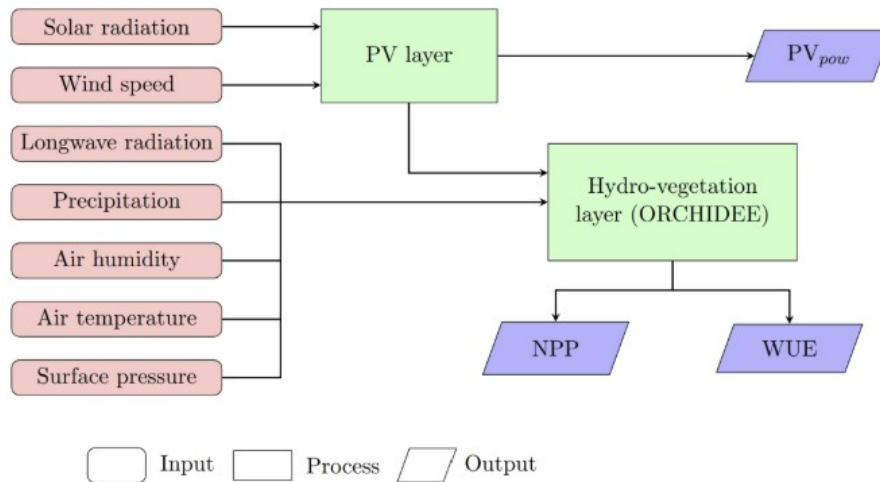


Figure 1: ORCHIDEE workflow, with examples of simulated data (NPP: Net Primary Production; WUE: Water Use Efficiency) produced from input climate data. The PV layer is the module developed by Rapella et al. [1].

By assigning ORCHIDEE parameters typical of agricultural soil, integrating the PV model into ORCHIDEE makes it possible to simulate the dynamics of agriPV ecosystems consistently. The model outputs provide diagnostics from which key indicators are calculated for assessing and analysing the performance of these agro-photovoltaic systems.

1.1.2 Initialisation of the ORCHIDEE model

We performed several simulations, modifying the parameters according to the issue being investigated. ORCHIDEE requires an initialisation stage in which its equations are brought into equilibrium. This is achieved in particular through “spin-ups”, which balance the equations governing the model parameters, a method commonly used in climate and ocean modelling.

We studied 12 configurations: one reference configuration consisting of crops without photovoltaic panels and with a nitrogen-input rate equal to 100% of the agricultural reference rate established by the Food and Agriculture Organization of the United Nations [10], denoted CTRL 100%N; and 11 configurations with solar panels, with nitrogen-input rates ranging from 0% to 100% of the reference rate in 10-percentage-point increments, denoted PV X%N, where X is the configuration’s nitrogen-input rate.

The simulations cover the Iberian Peninsula from 1991 through 2020. The Iberian Peninsula was selected because of its climatically interesting location. It contains several biomes but is predominantly arid in the south, a feature that is expected to become more pronounced over time as temperatures and drought increase in this geographical area because of the climate crisis [11].

The simulations were performed for two types of crops that are both actually grown on the Iberian Peninsula: C3 crops, which form molecules containing three carbon atoms during photosynthesis, such as soybean and wheat; and C4 crops, which form molecules containing four carbon atoms, such as maize and sugar cane. According to the literature, C4 crops are more drought-resistant than C3 crops [12]. Once the simulations had been completed, we used the architecture of the ORCHIDEE output data to access their results.

We then applied a filter that retained only geographical areas associated with relatively high agricultural production, thereby refining the results. At least 10% of each retained geographical area had to be devoted to agriculture. The resulting mask assigns either 0 or 1 to each point on the map to indicate whether or not it is included. Its purpose is to exclude geographical areas where agriculture is not feasible, such as the city centres of Madrid and Barcelona and the Pyrenees.

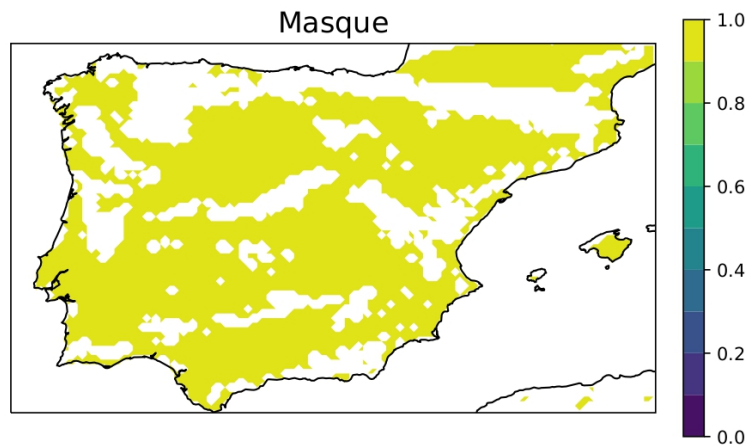


Figure 2: Geographical mask applied to Spain. Only agricultural areas are retained in order to refine the values obtained from the analysis.

1.2 Limitations of the PV model

The photovoltaic-panel simulation model is primarily based on a study of wind flow through solar-panel fields (Macdonald et al. [8]). It models wind flow as a function of parameters including wind speed and panel height, treating it as a flow through an urban-building canopy. Each solar panel is represented as a solid block. Although the theoretical model contains numerous variables, we focus on two that are included in the simulation: wind speed well above the field, where it is unaffected by changes at ground level, and solar-panel height. ORCHIDEE itself only accounts for wind 1.2 m above the ground; the wind far above the field concerns only the PV model. For low panel heights ($H \leq 3.5$ m), the current simulation uses the theoretical model and takes the mean wind speed between the ground and the panels as the speed experienced by the plants.

To assess the model's accuracy, we obtained wind-speed measurements from an agriPV field and compared them with the theoretical model on which our solar-panel simulator is based. The measurements were taken at SIRTa, an experimental facility in Palaiseau, as part of research conducted independently of ours. We compared the wind-flow speed measured under the solar panels with that simulated by the PV model for a given wind speed at a height of 10 m, taking into account all the site-specific parameters provided to us, including panel height, spacing, size, and angle.

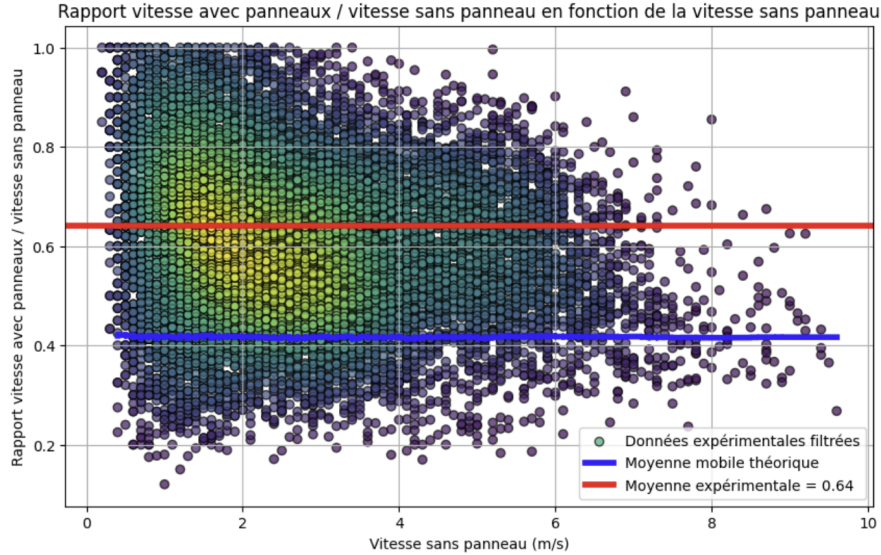


Figure 3: Ratio between wind speed at 10 m and wind speed between the solar panels, for the measurements (red line) and the simulation (blue line).

There is a noticeable difference between the theoretical model and the measurements. This difference may have several origins. For example, the measurements may not be perfectly representative because the field from which they were obtained is experimental. Nevertheless, we critically examine the model in order to identify ways of making it more representative of reality, and reconsider the assumptions underlying the chosen simulation model.

First, the theoretical study proposes a model whose profile varies with solar-panel height. The current simulation assumes that the panels are installed at a height $H \leq 3.5$ m [1]. This assumption matters because the profiles predicted for different panel heights differ significantly; moving from one regime to another changes the wind speed experienced by the plants in a way that could alter the results.

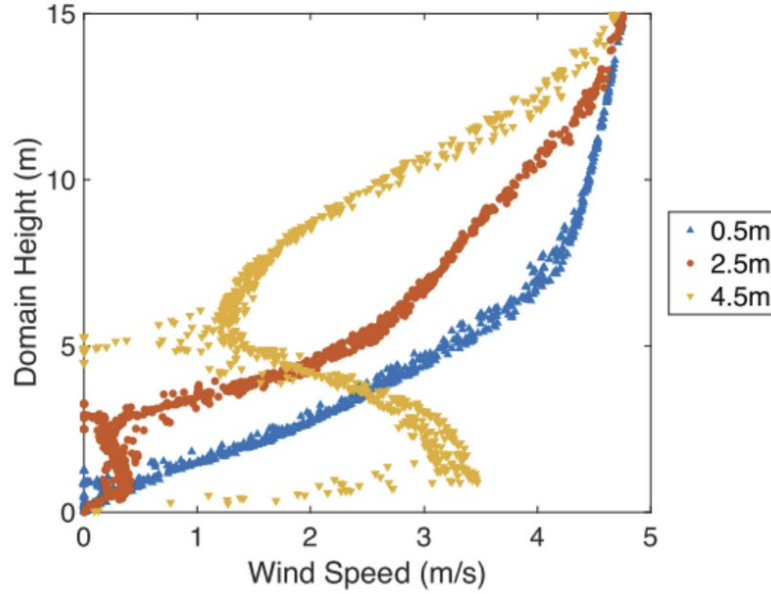


Figure 4: Wind-flow profiles as a function of solar-panel height, according to Williams et al. [9].

The graphs above show that the model of Williams et al. [9] predicts faster wind flow under solar panels when the panels are higher, in this case 4.5 m. The plants would therefore experience a higher wind speed, affecting key parameters such as temperature and evapotranspiration.

We believe that the theoretical model could be improved to represent reality more faithfully. One option would be to move away from the urban-canopy model and towards a model closer to tree canopies. A first model with two porous layers could be used to represent wind flow theoretically. It would contain a dense layer representing the solar panels and a second, less dense layer representing the plant-filled space between the ground and the panels. The model could be further improved by increasing the number of layers and accounting for the fact that vegetation becomes increasingly dense closer to the ground. Ultimately, a single porous layer beneath the panels could be considered, with density following a downward vertical gradient.

2 Results

We assess the relevance of agriPV farms using conventional indicators derived from the output of the ORCHIDEE model. These indicators describe productivity, nitrogen-input use, plant water use, and greenhouse-gas emissions from crops.

2.1 Influence on agricultural production

To measure the influence of photovoltaic panels and nitrogen inputs on food production, we study net primary production (NPP), which measures the production of plant biomass. It is therefore desirable to maximise this value.

To obtain an overall view of how the configurations affect production, we examine mean NPP aggregated over one year for each configuration and for C3 and C4 crops.

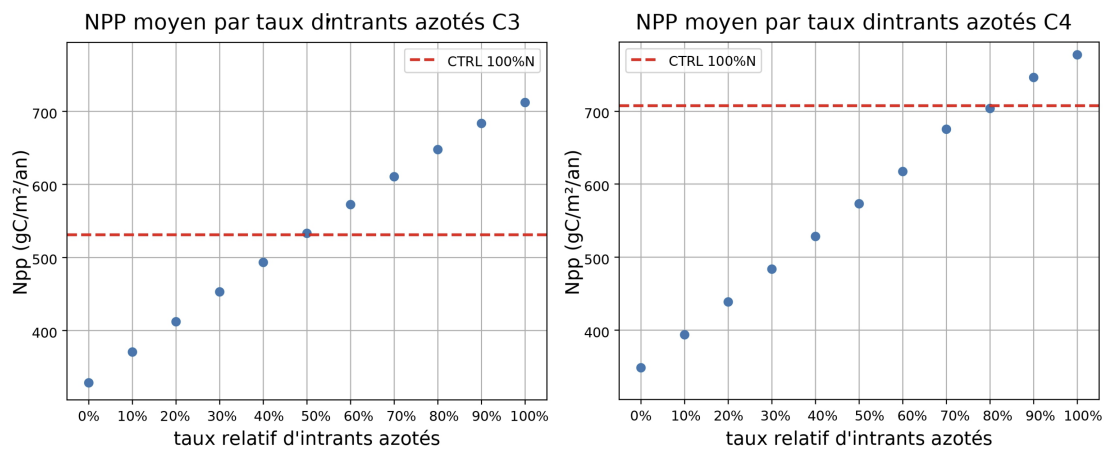


Figure 5: Mean production ($\text{gC/m}^2/\text{year}$), averaged over the thirty years from 1991 through 2020, as a function of the relative nitrogen-input rate in the PV configurations. The red line shows mean production in the reference configuration. Left: C3 crops. Right: C4 crops.

For C3 crops, the NPP trend in Figure 5 reveals a linear dependence of agricultural production on the quantity of nitrogen inputs. This first result is unsurprising: a plant can reasonably be expected to produce more organic matter when more nutrients are available. The graph nevertheless shows a second result. NPP in the reference configuration, without photovoltaic panels, is 530 grams of carbon per square metre per year, 25% lower than NPP in the configuration with photovoltaic panels and 100% nitrogen inputs, which reaches 710 grams of carbon per square metre per year. Production in the reference configuration is at the same level as NPP in the PV configuration with 50% nitrogen inputs.

Thus, for the same quantity of nitrogen inputs in the soil, agricultural production is greater on an agrivoltaic farm than for crops without panels. Alternatively, if the objective is not to increase production, installing solar panels over the crops makes it possible to reduce the quantity of nitrogen inputs while obtaining the same agricultural yield.

For C4 crops, Figure 5 shows that NPP in the reference configuration is 710 grams of carbon per square metre per year, compared with 880 grams in the PV configuration with 100% nitrogen inputs. Production in the reference configuration is therefore 20% lower than with photovoltaic panels. Reference production is at the same level as NPP in the PV configuration with 80% nitrogen inputs. Comparing the C3 and C4 reference configurations also shows that C4 production exceeds C3 production. This is consistent with Spain's semi-arid climate, since C4 crops are more drought-resistant than C3 crops.

The configuration with photovoltaic panels therefore produces a greater improvement in agricultural yields for C3 crops than for C4 crops, as well as a larger reduction in the nitrogen inputs needed to obtain the same production value.

This aggregated result does not, however, reveal the specific year-to-year production behaviour of each configuration. Examining the change in production over time shows both its variability and its stability. We therefore use a heat map whose horizontal axis is the relative nitrogen-input rate, whose vertical axis covers the period from 1991 through 2020, and in which the colour of each cell represents NPP in $\text{gC}/\text{m}^2/\text{year}$.

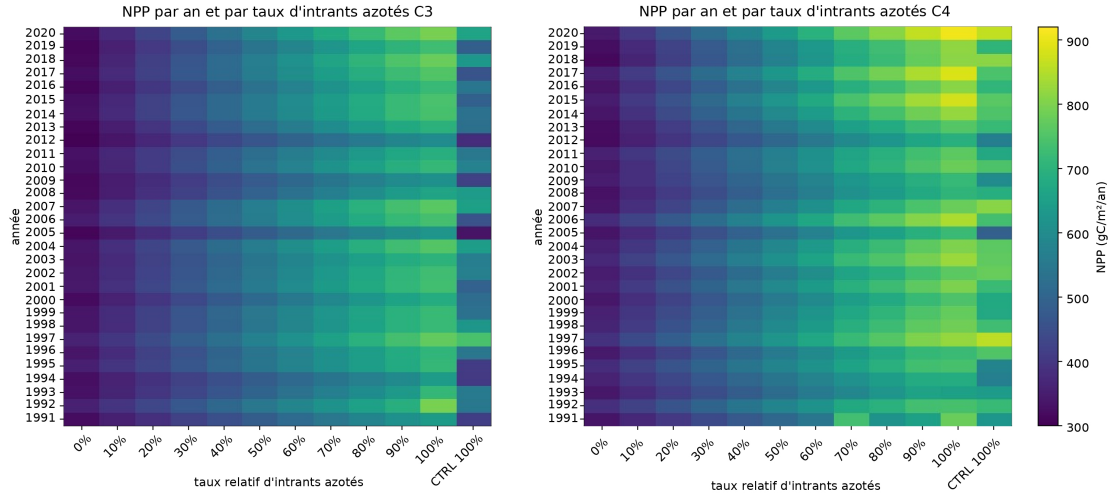


Figure 6: Mean annual production ($\text{gC}/\text{m}^2/\text{year}$) from 1991 through 2020 as a function of the relative nitrogen-input rate in the configurations with photovoltaic panels and in the reference configuration (CTRL 100%). Left: C3 crops. Right: C4 crops.

The first observation is the instability of annual production in the C3 reference configuration. Across the thirty years studied, there appear to be “good” and “bad” agricultural years. The reference configuration is fairly dependent on the conditions specific to each year, whereas this phenomenon appears less pronounced in the configurations with photovoltaic panels. This suggests that panels increase crop resilience. Production instability is also apparent in the C4 reference configuration, although the contrast with PV configurations having high relative nitrogen-input rates is less pronounced.

We describe this production variability in greater detail by considering within-year NPP variation between the most productive month and the month with the lowest production. The following graph shows the difference in NPP between those two months for each year and the evolution of that difference from 1991 through 2020.

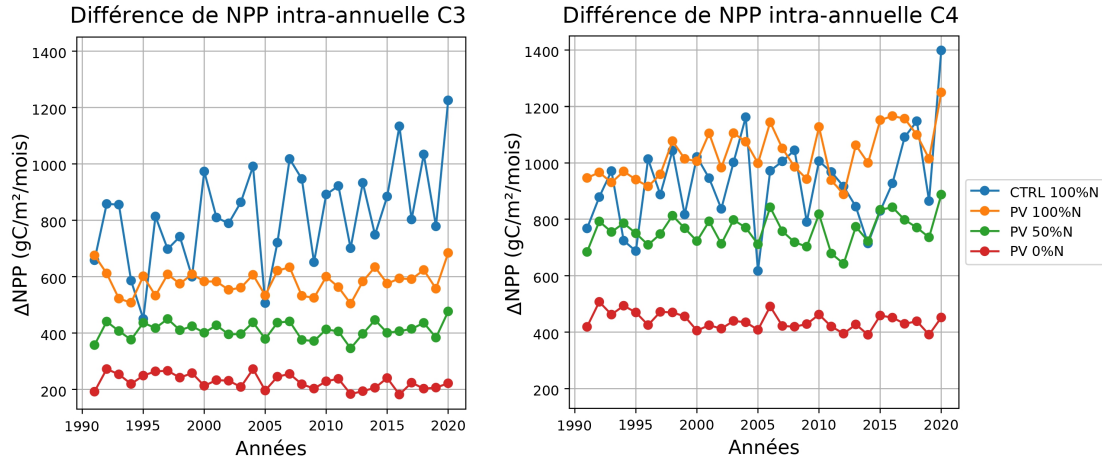


Figure 7: Within-year and between-year production variations from 1991 through 2020 ($\text{gC/m}^2/\text{month}$), shown for the reference configuration (CTRL 100%N) and for PV configurations with relative nitrogen-input rates of 0%, 50%, and 100%. Left: C3 crops. Right: C4 crops.

For C3 crops, the within-year difference between the most productive and least productive months averages $800 \text{ gC/m}^2/\text{month}$ in the reference configuration, compared with $400 \text{ gC/m}^2/\text{month}$ in the PV 50%N configuration, which has equivalent agricultural production according to Figure 7. In 2020, the variation between the highest- and lowest-production months reaches $1,200 \text{ gC/m}^2/\text{month}$ in the CTRL 100%N configuration. In the same year, although the PV 100%N configuration has higher production, its within-year variation is lower, at $700 \text{ gC/m}^2/\text{month}$.

Furthermore, within-year variation remains relatively constant across years in the PV configurations: about $600 \text{ gC/m}^2/\text{month}$ at a relative nitrogen-input rate of 100%, 400 at 50%, and 200 at 0%. By contrast, within-year production variation in the reference configuration changes sharply from one year to another, ranging from 440 to $1,200 \text{ gC/m}^2/\text{month}$.

Thus, agriPV farms growing C3 crops not only increase production for the same quantity of nitrogen inputs, but also provide more reliable seasonal and annual yields.

The influence of photovoltaic panels on within-year production variation is less pronounced for C4 crops. Although these variations are more stable from one year to another in the PV configurations than in the reference configuration, their values are similar for PV 100%N and CTRL 100%N. This indicates substantial seasonal production instability in both configurations. Photovoltaic panels are therefore effective in reducing between-year and between-month production variations for C3 crops, whereas the effect is not as pronounced for C4 crops.

It is also relevant to examine the geographical distribution of agricultural production in the CTRL 100%N and PV 100%N configurations using the following maps, based on annual mean NPP.

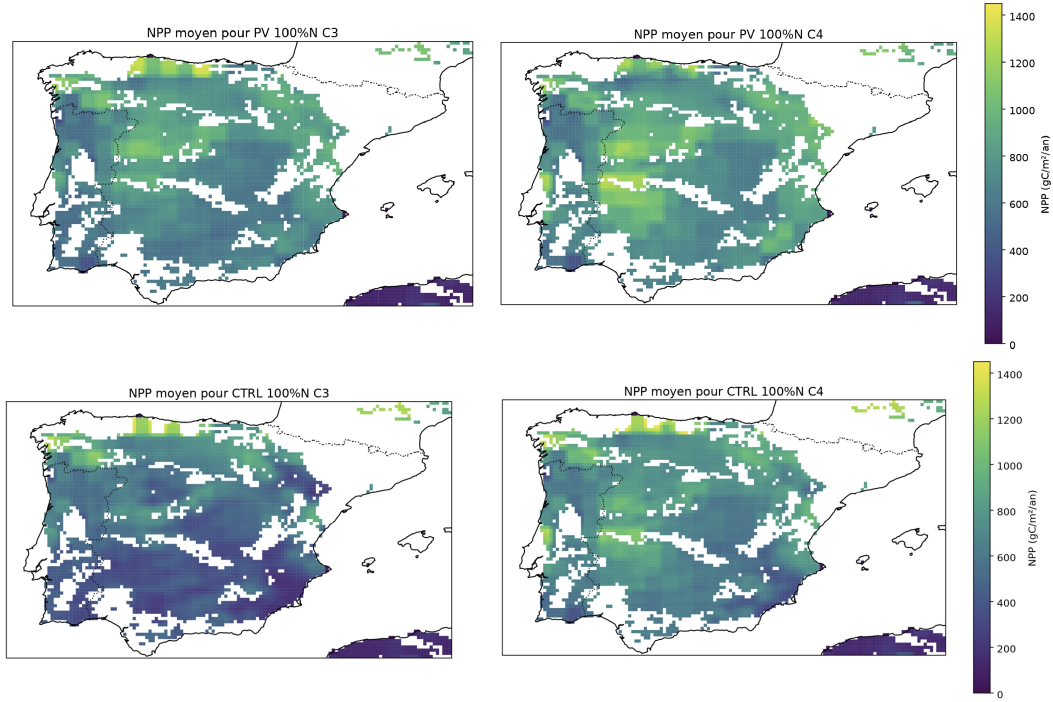


Figure 8: Maps of mean crop production ($\text{gC/m}^2/\text{year}$), averaged over the thirty years from 1991 through 2020, in the reference configuration (top) and in PV configurations with a relative nitrogen-input rate of 100% (bottom). Left: C3 crops. Right: C4 crops.

Figure 8 shows that production is much more spatially homogeneous in the PV 100%N configuration than in the reference configuration, for both C3 and C4 crops. Production values are also substantially higher with photovoltaic panels than in the reference configuration for C3 crops, particularly in a clearly defined area northeast of the Portuguese border. The production difference is also noticeable for C4 crops in the same geographical area. C4 crops additionally have higher production than C3 crops in the same configuration, consistently with the greater resistance of C4 crops to Spain’s semi-arid climate mentioned above.

Another useful indicator for studying the influence of photovoltaic panels on plant growth and agricultural production is mean plant leaf area. We therefore examine the Leaf Area Index (LAI). This index also reflects plant health: a high LAI is associated with thriving, healthy crops. We consequently seek to maximise LAI.

The following graph shows the evolution of mean LAI over a year in the CTRL 100%N, PV 100%N, PV 50%N, and PV 0%N configurations for C3 and C4 crops.

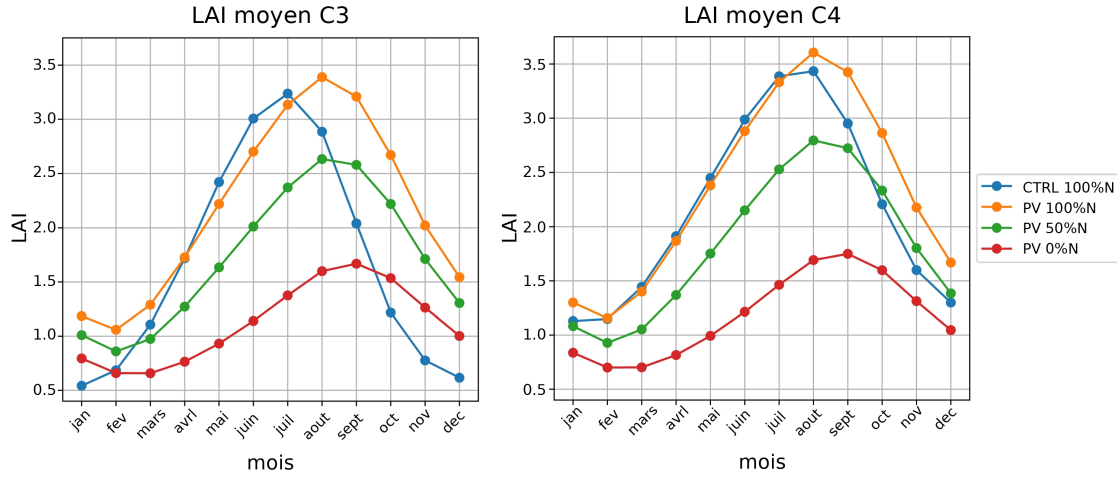


Figure 9: Leaf Area Index, averaged over thirty years from 1991 through 2020, by month in the PV configurations (0%, 50%, and 100%) and in the reference configuration. Left: C3 crops. Right: C4 crops. LAI is dimensionless.

The first observation is the seasonality of leaf area, which develops during the first part of the year and decreases during the second. For C3 crops, however, maximum leaf growth occurs later in the summer in the PV configurations than in the reference configuration. Depending on the nitrogen-input rate, it occurs in August or September with panels, compared with July in the reference configuration. LAI is also maintained more effectively at the end of the year in the PV configurations. For example, LAI in PV 50%N is higher than the reference LAI from September through February, or for half of the year. These results are less visible for C4 crops.

For C3 crops, LAI is higher for most of the year in PV 100%N than in the reference configuration, except in May, June, and July, when the values are very close. This result again applies less clearly to C4 crops, for which PV 100%N and CTRL 100%N exhibit fairly similar behaviour.

These graphs therefore show that agriPV improves the health of C3 plants for most of the year at the same nitrogen-input rate, and also maintains plant health more effectively in late summer and autumn than a conventional agricultural system.

Ultimately, this plant-health result supports our earlier NPP finding: agriPV increases production by spreading plant growth over a longer period. It can even reduce nitrogen-input rates while maintaining the same production. These effects are more pronounced for C3 crops than for C4 crops. To understand them more fully, we now examine how plants use nitrogen inputs in PV agriculture.

2.2 Influence on nitrogen-use efficiency

To study the effect of the different nitrogen-input rates in an agriPV system relative to a conventional agricultural system, we consider how efficiently plants use nitrogen in the different configurations. We use the Nitrogen Use Efficiency (NUE) index. A high NUE value indicates optimised use of nitrogen fertiliser resources and therefore less relative soil pollution.

We begin by studying annual NUE variation in the reference configuration and in PV configurations with nitrogen-input rates of 100%, 50%, and 0%.

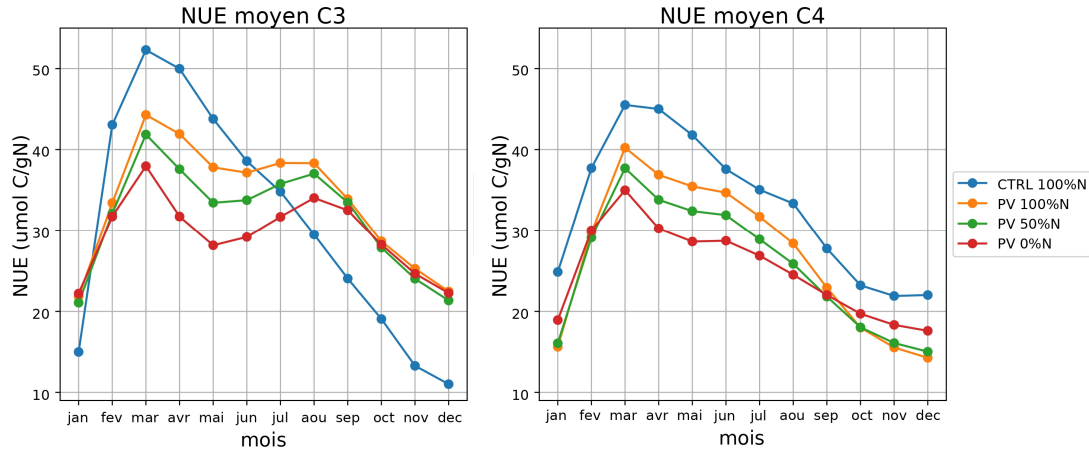


Figure 10: Nitrogen-use efficiency ($\mu\text{molC/gN}$), averaged over thirty years, by month in the PV configurations (0%, 50%, and 100%) and in the reference configuration. Left: C3 crops. Right: C4 crops.

For C3 crops, NUE in the reference configuration is initially higher than NUE in the PV configurations during the first half of the year. It then becomes lower than NUE in the PV configurations during the second half. The graph also shows that nitrogen-use efficiency decreases as the nitrogen-input rate increases throughout the year.

For C4 crops, NUE in the reference configuration is higher than in the PV configurations throughout the year. In terms of nitrogen-use efficiency, agriPV is therefore detrimental to C4 crops.

The C3 results are mixed. A more constant NUE over the year may support greater annual production by avoiding a short-lived period of intense plant growth. Nevertheless, because the relative benefits of the PV configurations compared with the reference configuration vary sharply between months, it is difficult to estimate the NUE benefit of agriPV month by month. We therefore examine NUE aggregated over one year and averaged over thirty years for the different C3 and C4 configurations.

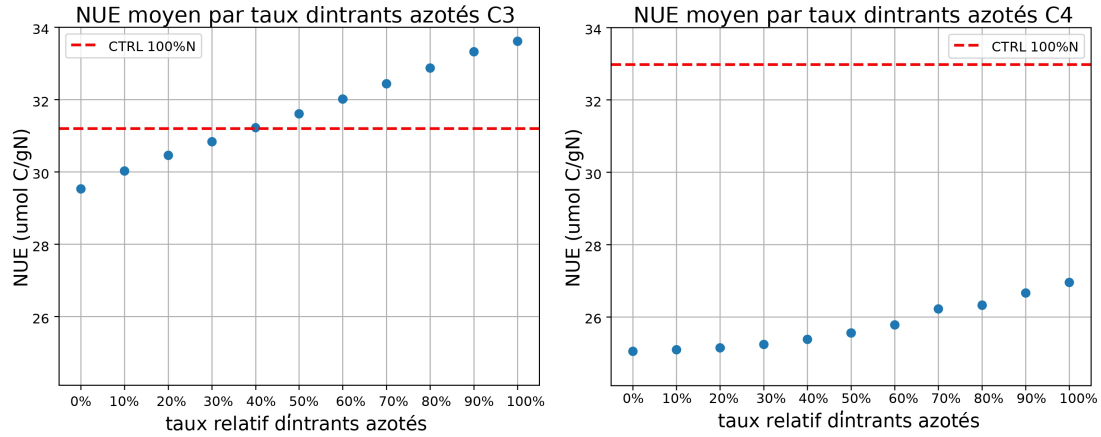


Figure 11: Nitrogen-use efficiency as a function of the relative nitrogen-input rate in the PV configurations. The red line gives nitrogen-use efficiency in the reference configuration. Left: C3 crops. Right: C4 crops.

The C4 graph confirms the previous result: agriPV reduces nitrogen-use efficiency relative to an agricultural system without photovoltaic panels, regardless of the nitrogen-input rate. The reduction is substantial, with an approximately 18% decrease in NUE between CTRL 100%N and PV 100%N.

The C3 graph again shows that NUE decreases as the nitrogen-input rate increases. More importantly, annual aggregated NUE is higher with panels than in the reference configuration at the same nitrogen-input rate. The same nitrogen-use efficiency as in the reference configuration is even achieved with only 40% nitrogen in an agriPV system.

In conclusion, agriPV increases nitrogen-use efficiency for C3 crops and has the opposite effect for C4 crops. The optimisation of nitrogen resources observed for C3 crops is relevant in the context of reducing soil pollution and limiting resource use. With the same objective, we now examine the optimisation of water resources.

2.3 Influence on the use of water resources

To study the optimisation of water resources in agrivoltaic systems, we consider two indicators. First, we examine plant water stress using the Water Stress Index (WSI), followed by plant Water Use Efficiency (WUE).

WSI is particularly useful for demonstrating how photovoltaic panels influence crop water conditions. It is therefore relevant to the study of agrivoltaics in a context of increasing drought on the Iberian Peninsula due to climate change.

WUE measures the efficiency with which a plant uses water to produce biomass or fix carbon. It is the ratio between the amount of carbon assimilated by the plant through photosynthesis, or Gross Primary Production, and plant evapotranspiration, consisting of transpiration plus evaporation. This quantity is useful for quantifying plant adaptation to water stress because an increase in WUE can reflect a strategy for maximising carbon assimilation under water-limited conditions.

We first show the mean annual evolution of WSI in the reference configuration and the PV 100%N, PV 50%N, and PV 0%N configurations for C3 and C4 crops.

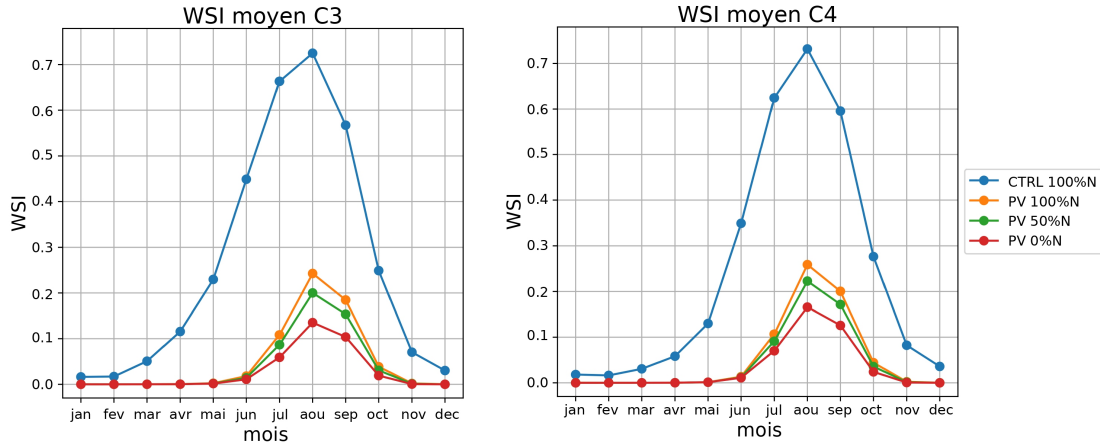


Figure 12: Crop water stress, averaged over thirty years, by month in the PV configurations (0%, 50%, and 100%) and in the reference configuration. Left: C3 crops. Right: C4 crops. WSI is dimensionless.

Both graphs show that plants experience much greater stress in the reference configuration, where WSI reaches 0.72, than in the other configurations, where WSI is 0.24, 0.20, and 0.13 for PV 100%N, PV 50%N, and PV 0%N respectively. More water is therefore available to crops with solar panels than to crops without them. This may explain some of the results observed in Section 2.1, where summer organic production and leaf growth were lower without solar panels than in the PV configurations.

The most plausible hypothesis is that shade from the panels reduces plant evapotranspiration, thereby promoting growth over a longer period.

The two graphs also show a correlation between nitrogen quantity and water stress: the lower the nitrogen-input rate, the lower the water stress. This may be explained by the production of less plant biomass, resulting in less stress for the same amount of available water. Finally, the graphs show that C3 and C4 crops behave similarly under water stress: both respond positively, and to a similar extent, to the presence of photovoltaic panels.

Plants under solar panels therefore withstand heat and water tension more effectively and grow for longer. This suggests a greater capacity to adapt to climate change. From a resource perspective, this capacity is essential because it makes better use of agricultural resources such as water and nitrogen inputs, which will become scarcer or more expensive in the future.

To study the capacity of plants to optimise their water use in agriPV systems, we examine annual WUE averaged over thirty years in the different C3 and C4 configurations.

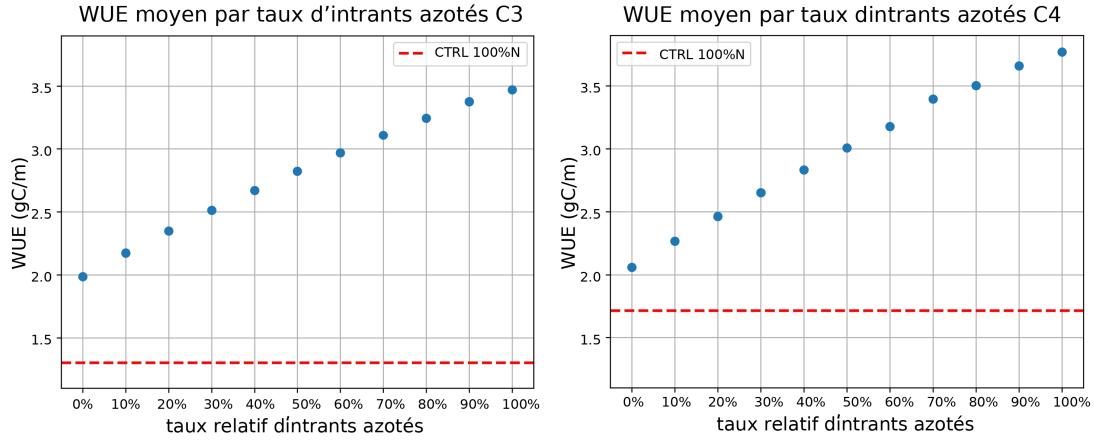


Figure 13: Water-use efficiency (gC/m), averaged over the thirty years from 1991 through 2020, as a function of the relative nitrogen-input rate in the PV configurations. The red line gives water-use efficiency in the reference configuration. Left: C3 crops. Right: C4 crops.

Both graphs show that the available water is used more efficiently in every configuration with solar panels than in the reference configuration. This may be due to the greater leaf area maintained throughout the year in the PV configurations, which promotes optimised water use over the entire year. By contrast, the reference configuration, in which a large leaf area is present only for a short period, creates inefficiency.

Water-use efficiency also increases with the nitrogen-input rate. Once again, because leaves are present in larger quantities at high nitrogen rates, the previous hypothesis is consistent with water use being optimised in these configurations.

AgriPV therefore benefits water-use efficiency for both C3 and C4 crops, although the benefit is slightly greater for C3 crops. The difference between the two crop types may result from the better adaptation of C4 crops to arid climates. By covering the plants, the panels reduce both incident radiation and wind speed, which may make the environment around the plants less arid. C3 crops, which are more sensitive to such conditions, would benefit more than C4 crops.

Beyond the impact of agriPV on resources such as nitrogen and water, we now consider greenhouse-gas emissions from an agriPV system.

2.4 Influence on greenhouse-gas emissions

As discussed in Section 2.2, photovoltaic panels improve nitrogen-use efficiency. Better plant use of nitrogen implies that, in configurations with panels, a larger proportion of the nitrogen inputs is consumed by the plants and a smaller proportion remains in their environment. This could be a significant result because, as discussed in the introduction, excess nitrogen in soils pollutes the environment and releases greenhouse gases such as nitrous oxide, hereafter denoted N_2O .

We therefore consider the quantity of N_2O emitted by the crops in the different configurations. First, we show mean monthly N_2O emissions between 1991 and 2020 in the reference configuration and in three configurations with panels.

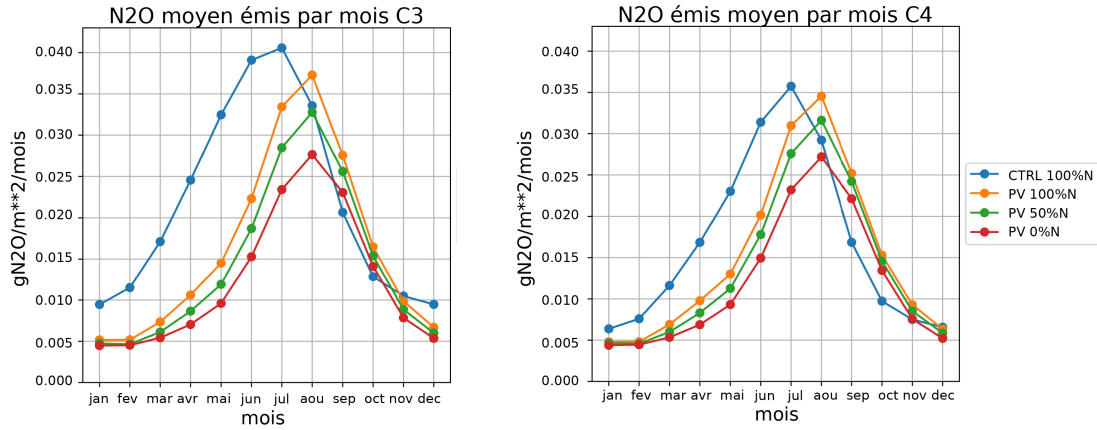


Figure 14: Monthly N₂O emissions (gN₂O/m²/month), averaged over all years from 1991 through 2020, in the CTRL 100%N, PV 100%N, PV 50%N, and PV 0%N configurations for C3 and C4 crops.

Regardless of the configuration or crop type, plants release the most N₂O during the summer. This result is explained by higher summer temperatures, which increase N₂O release from soils. Throughout the first half of the year, N₂O emissions in the reference configuration are substantially higher than in the configurations with panels. During the second half, they become slightly lower than in the PV configurations.

Throughout the year, N₂O emissions are also lower when the nitrogen-input rate in the soil is lower. This is consistent with the fact that N₂O originates from the nitrogen contained in the soil.

Comparing the C3 and C4 graphs shows that C4 crops emit less N₂O in all configurations. More importantly, the reduction in N₂O emissions between the reference configuration and the PV configurations is smaller for C4 crops than for C3 crops. AgriPV therefore appears to have a greater influence on C3 crops in terms of N₂O emissions, although the panels do also reduce emissions from C4 crops.

To obtain a clearer picture of the differences between configurations, we plot mean annual N₂O emissions from 1991 through 2020 against the nitrogen-input rate.

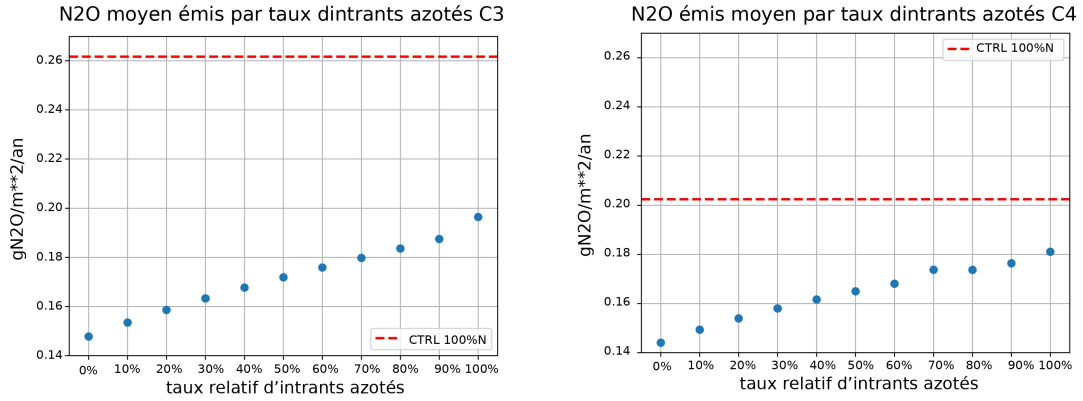


Figure 15: Annual N_2O emissions ($\text{gN}_2\text{O}/\text{m}^2/\text{year}$), averaged over the thirty years from 1991 through 2020, as a function of the relative nitrogen-input rate in the PV configurations. The red line gives the value for the reference configuration. Left: C3 crops. Right: C4 crops.

For both crop types, N_2O emissions in the reference configuration are substantially higher than in the PV configurations. Comparing this graph with Figure 5 shows that, for C3 crops in the reference configuration, the ratio of annual production to annual N_2O emissions, $\text{NPP}/\text{N}_2\text{O}$ in $\text{gC}/\text{gN}_2\text{O}$, is approximately 2,040 $\text{gC}/\text{gN}_2\text{O}$. In PV 100%N, the same ratio is approximately 3,500 $\text{gC}/\text{gN}_2\text{O}$. The panels therefore very clearly provide greater agricultural production for the same N_2O emissions.

Within the PV configurations, N_2O emissions increase with the nitrogen-input rate, again consistently with the process through which N_2O is generated. The change in N_2O emissions with the nitrogen-input rate also appears linear, meaning that changing the quantity of nitrogen inputs supplied to the crops does not in itself provide an additional benefit. Using Figure 5 to calculate annual $\text{NPP}/\text{N}_2\text{O}$ even shows that production per unit of emitted N_2O decreases as the nitrogen-input rate decreases, linearly from 3,500 $\text{gC}/\text{gN}_2\text{O}$ for PV 100%N to 2,200 $\text{gC}/\text{gN}_2\text{O}$ for PV 0%N. Reducing nitrogen inputs is therefore less advantageous when assessed in terms of the organic matter produced for the same N_2O emissions.

N_2O emissions are higher for C3 crops than for C4 crops across all configurations. More importantly, the decrease in emissions is much greater for C3 crops, with a 23% reduction from CTRL 100%N to PV 100%N, than for C4 crops, for which the reduction is 10%. This again shows that agriPV is more advantageous for C3 than C4 crops in terms of N_2O emissions because C3 crops derive a larger relative benefit.

Overall, the preceding graphs show that installing photovoltaic panels above crops substantially reduces the quantity of N_2O released into the atmosphere by agriculture for the same quantity of organic matter produced.

This result can be considered alongside the earlier NUE analysis. Section 2.2 showed that C3 crops use nitrogen more efficiently in agriPV systems than in conventional agricultural systems. With panels, the plants consume a greater proportion of the nitrogen inputs, which contribute more to their growth than in the configuration without panels. The soil in an agriPV system therefore contains fewer nitrogen inputs that have not been used by the plants and consequently releases fewer nitrogen compounds such as N_2O . The NUE results thus partly explain the N_2O -emission benefits of agriPV.

Finally, the results in Sections 2.1 through 2.4 show that agrivoltaics is more relevant to C3 than C4 crop production. At the same nitrogen-input rate, agriPV improves all the indicators studied for both C3 and C4 crops, except NUE for C4 crops. The improvements are nevertheless consistently larger for C3 crops, which are more sensitive to arid conditions and therefore benefit more from the cover provided by photovoltaic panels. In the remainder of the study, we consequently consider only C3 crops.

3 Extreme climatic conditions

The results in Section 2.1 show that production can fall in certain years, which generally correspond to drought years.

The literature identifies four years among the past thirty years with a low SPI-12 value, or Standardized Precipitation Index (Catalogue of drought events in peninsular Spanish along 1916–2020 period, Trullenque-Blanco et al. [7]). Those years are 1994, 1995, 2005, and 2012.

Studying the precipitation data for Spain over the past thirty years gives the following graph.

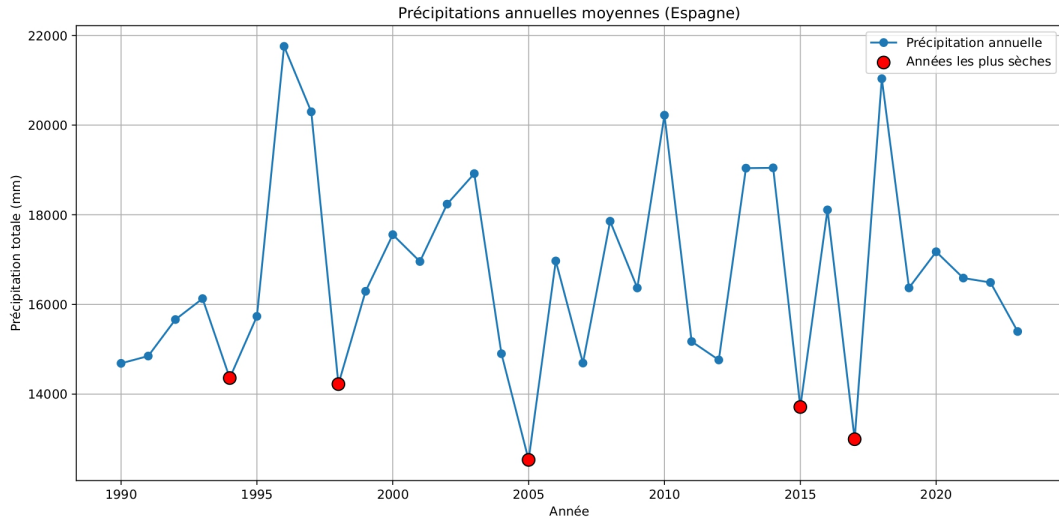


Figure 16: Mean annual precipitation (mm) in Spain from 1991 through 2023. The five years with the lowest precipitation among those shown are highlighted in red.

The five years with the lowest precipitation appear in red. In increasing order, they are 2005, 2017, 2015, 1998, and 1994. To corroborate the effect of these dry conditions on agricultural production, we use the following graph.

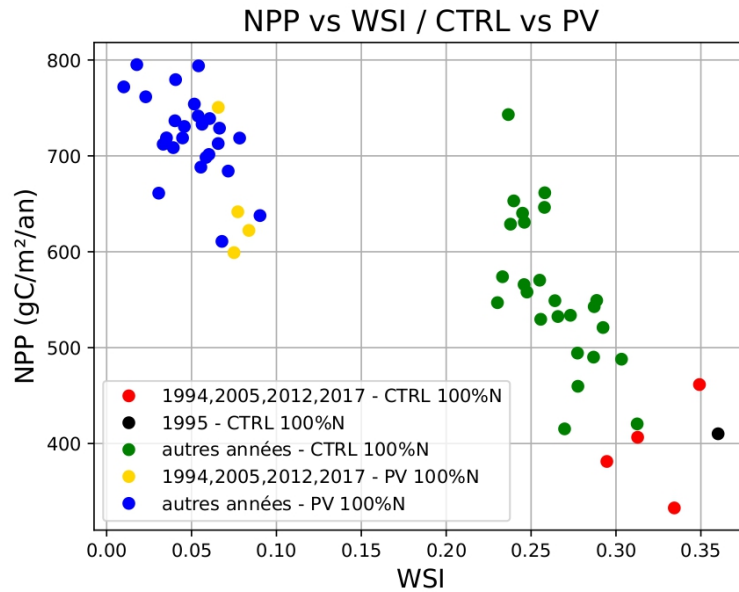


Figure 17: Years from 1991 through 2020 plotted according to their NPP and WSI values in the CTRL 100%N and PV 100%N configurations.

The graph shows every year from 1991 through 2020 according to NPP and WSI in the CTRL 100%N and PV 100%N configurations.

Examining the years in CTRL 100%N, shown as green, red, and black points, reveals that 1994, 2005, 2012, and 2017, shown in red, together with 1995, shown in black, combine the highest water stress with the lowest production. This confirms the extreme nature of 1994, 2005, 2012, and 2017 identified above from climate data and the literature.

The targeted analysis that follows therefore focuses on 1994, 2005, 2012, and 2017, hereafter referred to as the “dry years”. We excluded 1995 because 1994 may have had a substantial influence on the following year’s climate conditions. In the study by Trullenque-Blanco et al., the 1994–1995 drought is divided between half of each year [7]. We consequently count only 1994 as a drought episode. Likewise, on the basis of the precipitation data, 1998 could have been included under the same criteria if we had chosen a larger number of dry years.

The preceding graph first demonstrates the influence of photovoltaic panels on crops during dry years. In terms of both water stress and production, PV 100%N performs substantially better overall than the reference configuration. The result also applies to the dry years, shown in yellow: their water stress is lower than in every year of the reference configuration, and their production exceeds that of 76% of the reference years.

In addition, the production gap between dry years and other years is much smaller with PV than in the reference configuration. The range is $200 \text{ gC/m}^2/\text{year}$ in the former and approximately $400 \text{ gC/m}^2/\text{year}$ in the latter. This suggests greater production stability and therefore a lower risk of shortages.

We now examine crop behaviour over the course of these dry years. First, we plot mean annual LAI in the reference configuration and in PV configurations with nitrogen-input rates of 100%, 50%, and 0%, both for the drought years and for all years from 1991 through 2020.

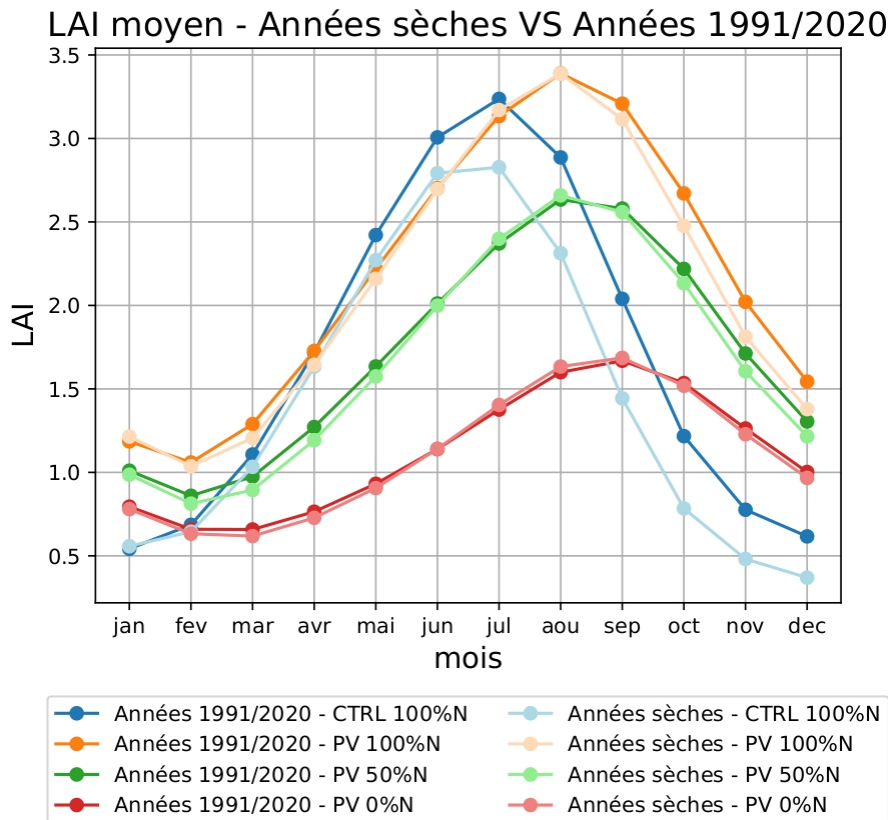


Figure 18: Mean annual evolution of LAI during the drought years and during all years from 1991 through 2020, for the CTRL 100%N, PV 100%N, PV 50%N, and PV 0%N configurations.

In the PV configurations, LAI is almost identical throughout the year in the dry-year average and in the 1991–2020 average. It is slightly higher in summer during the dry years and slightly lower during the rest of the year. By contrast, the reference configuration shows a noticeable difference in leaf area, with a sharp decrease under dry conditions. Since LAI reflects plant health, we conclude that photovoltaic panels allow plants to grow despite extreme climatic conditions.

In the reference configuration, LAI is lower throughout a dry year than in the 1991–2020 average, with the difference particularly pronounced in summer. Crops therefore appear to be in poorer health during dry years than over the complete 1991–2020 period.

Finally, comparing CTRL 100%N with PV 100%N confirms the finding from Section 2.1: LAI is higher, and plant health therefore better, in the PV configuration than in the reference configuration. We previously explained this difference by the improved water management in the agriPV system, which retains water in the crops for longer during the year. This improved management may also explain the good LAI performance of agriPV during dry years.

To investigate this question, we examine crop water stress during dry years. In order to highlight the difference between dry years and an average year, we plot the difference between values for the dry years and values for the full 1991–2020 period. Applying this approach to WSI in the different configurations gives the following graph.

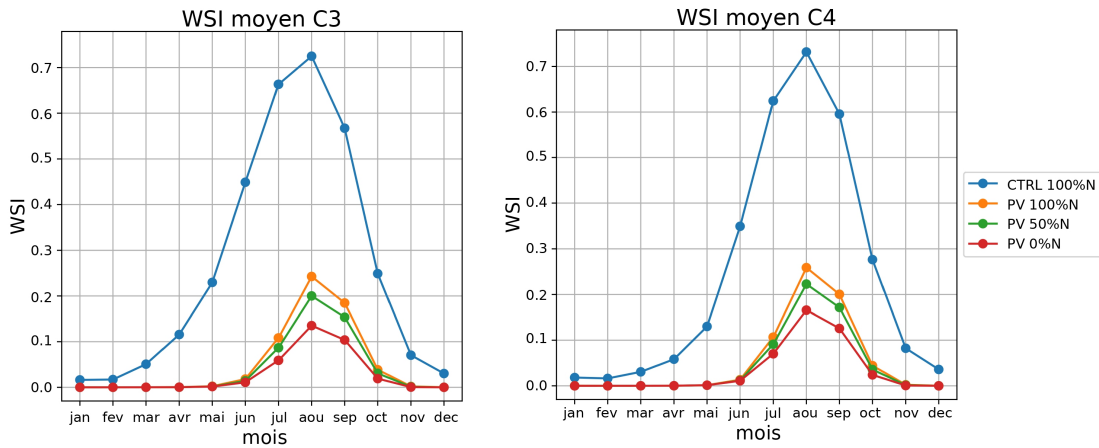


Figure 19: Mean monthly difference in WSI between the dry years and all years from 1991 through 2020, for the CTRL 100%N, PV 100%N, PV 50%N, and PV 0%N configurations.

Regardless of the configuration, water stress increases substantially during dry years. For CTRL 100%N, for example, annual mean WSI increases by 18%.

The increase in water stress does not, however, occur at the same time of year or with the same amplitude in the reference and PV configurations. It is larger and occurs earlier in the reference configuration. This shows that photovoltaic panels “cushion the drought”: they retain water in the crops more effectively during drought and therefore help the system withstand the water tension caused by extreme climatic conditions.

Beyond maintaining relatively stable plant health between an ordinary year and a dry year, the improved water management provided by agriPV may also maintain high production during dry years. We obtain this result by studying NPP, and in particular the change in production during dry years.

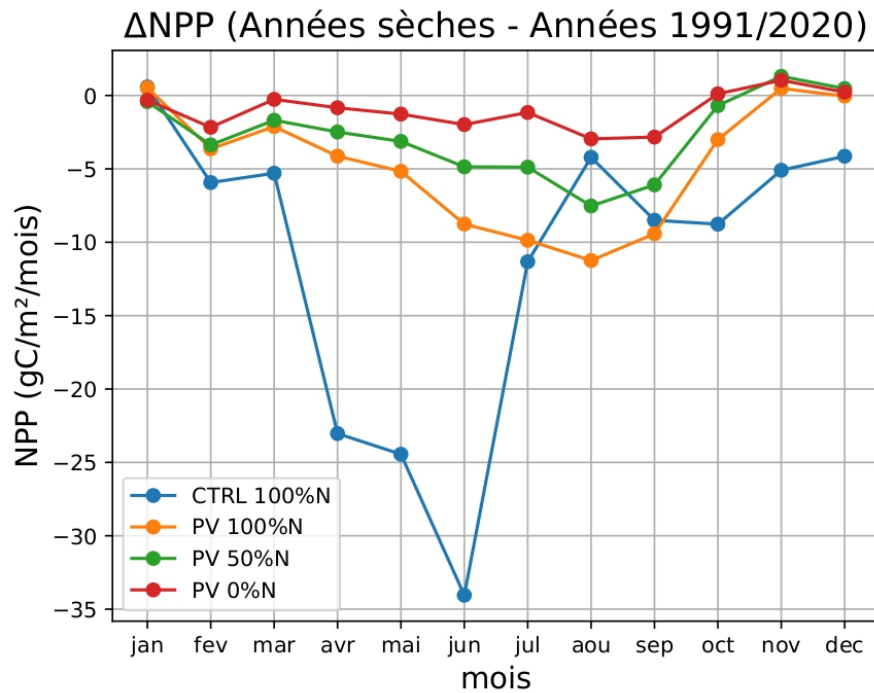


Figure 20: Mean monthly difference in NPP between the dry years and all years from 1991 through 2020, for the CTRL 100%N, PV 100%N, PV 50%N, and PV 0%N configurations.

Regardless of the configuration, crops again perform less well during a dry year than during an arbitrary year. NPP falls in almost every month in all configurations. The decrease is nevertheless much larger in the reference configuration, where it represents a 5–10% reduction in monthly production, than in PV 100%N, where it represents a 0.5–2% reduction. The decrease also occurs earlier in the reference configuration. In terms of production as well, photovoltaic panels therefore reduce the effect of drought and maintain more stable production between an ordinary year and a dry year.

Conclusion and discussion

In this study, we applied an agriPV model to the Iberian Peninsula in order to simulate crop behaviour and interactions with photovoltaic systems, and to assess the influence of an agriPV system on agricultural production and resource management.

The PV component of the model was developed by Rapella et al. [1]. Using experimental measurements taken at the SIRTa agriPV facility, we partially evaluated the model's accuracy. The model is largely satisfactory for this project, although we explored several potential improvements. It may be useful to increase the mean solar-panel height considered in the model so that it more closely reflects commonly encountered installations. A model closer to tree canopies, dividing the space between the panels and the ground into a greater number of layers, could also be investigated as a future improvement.

We analysed the impacts of agriPV on crop-ecosystem behaviour in terms of food production, the use of water and nitrogen resources, and fertiliser-related greenhouse-gas emissions. The numerical simulations show that, on the Iberian Peninsula and for C3 crops, agrivoltaic systems provide numerous benefits compared with conventional agricultural systems.

First, at the same nitrogen-input rate, agriPV provides a substantial increase in agricultural production, amounting to 33%, as well as an improvement in plant health. Several factors explain this result. The panels allow crops to retain water more effectively and consequently experience less water stress throughout the year. They also improve the use of water resources, as demonstrated by the increase in water-use efficiency in the presence of photovoltaic panels. This improved water management allows plants to continue developing during the summer, spreading production over the year, and maintains good plant health until the end of summer.

By comparison, production without photovoltaic panels is very intense in spring and early summer but then falls sharply as plant health deteriorates because water availability declines substantially. This explains the large annual difference in the quantity of organic matter produced by an agrivoltaic system and by a conventional agricultural system.

Second, agrivoltaics preserves water and nitrogen resources more effectively by improving the efficiency with which they are used. This result is accompanied by reductions in pollution and in greenhouse-gas emissions associated with excess nitrogen in soils.

Plant growth is distributed more evenly across the year in an agrivoltaic system, and the plants experience less water stress. Water and nitrogen inputs are therefore used more efficiently by crops whose growth is slower and more widely distributed over time than that of crops in a conventional agricultural system. AgriPV can reduce nitrogen inputs in the soil by up to 50% while maintaining the same agricultural production. Better use of nitrogen inputs supports greater growth, but also reduces the nitrogen left unused by plants in the soil and therefore the greenhouse gases generated from that nitrogen, such as N_2O , a greenhouse gas 273 times more powerful than CO_2 .

Third, agrivoltaics makes agricultural production more consistent from one year to another by providing greater resilience to extreme climatic events than a conventional agricultural system. The analysis restricted to drought years shows that the beneficial agriPV mechanisms described above, including improved management of water and nitrogen inputs, remain valid under extreme climatic conditions. This better resource management explains why agriPV limits production losses during dry years relative to a conventional agricultural system and preserves plant health. This resilience is particularly relevant because years with extreme climatic conditions will become more frequent as global warming progresses. Enabling the agricultural sector to maintain stable production with fewer shortages is therefore advantageous.

Finally, we extended the study to C4 crops. The benefits described above in terms of production, plant health, optimisation of water resources, and reduced N_2O emissions also apply to C4 crops in an agrivoltaic system. They are nevertheless less pronounced for C4 crops than for C3

crops. C4 crops even exhibit better nitrogen-use efficiency in fields without photovoltaic panels.

The difference between the benefits obtained for C3 and C4 crops may be due to the greater natural adaptation of C4 crops to an arid climate. The panels provide partial cover and therefore reduce both incident radiation and wind speed at canopy level. This microclimatic change is likely to reduce aridity in the immediate crop environment, particularly benefiting C3 crops, which are more sensitive to arid conditions than C4 crops.

Overall, our results show that agriPV can provide benefits in terms of agricultural production and reduced use of environmental resources. When the broader Water–Energy–Food–Ecosystem (WEFE) nexus is considered, the focus shifts from simply maximising production to balancing food production, resource use, and ecosystem health. The objective is to minimise environmental impacts such as water consumption, fertiliser use, and greenhouse-gas emissions while maintaining acceptable crop productivity, and agriPV responds well to this challenge. Our study nevertheless covered only a specific geographical area and a thirty-year period from 1991 through 2020. The overall results for agriPV are positive, but trade-offs must be found according to crop type, climatic conditions, and the level of fertiliser input.

Acknowledgements

We would like to express our sincere thanks to Philippe Drobinski, Jean-Marc Chomaz, The Laboratoire de Météorologie Dynamique (LMD), the E4C centre, and especially Lia Rapella for supervising us throughout this project.

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Glossary

PV Photovoltaic.

NPP ($\text{gC m}^{-2} \text{s}^{-1}$) Net Primary Production, a measure of plant biomass production intrinsically related to food production. It is measured in grams of carbon per square metre per second.

WUE (gC m^{-1}) Water Use Efficiency. It is calculated as the ratio between the amount of carbon assimilated by the plant through photosynthesis, or Gross Primary Production, and plant evapotranspiration, consisting of transpiration plus evaporation. It is measured in grams of carbon per metre.

WSI Water Stress Index, a dimensionless measure of plant water stress.

LAI Leaf Area Index, a dimensionless quantity describing the area of foliage present on a plant.

NUE ($\mu\text{mol C/gN}$) Nitrogen Use Efficiency, measured in micromoles of carbon per gram of nitrogen.

N₂O Nitrous oxide.

ORCHIDEE ORganising Carbon and Hydrology In Dynamic Ecosystems, a dynamic vegetation model designed to simulate complex interactions between climate and ecosystems.

C3 Crops that form molecules containing three carbon atoms during photosynthesis, including soybean and wheat.

C4 Crops that form molecules containing four carbon atoms during photosynthesis. These crops are more drought-resistant and include maize and sugar cane.

SPI Standardized Precipitation Index, used to identify drought periods from precipitation levels over a given period. It is calculated by normalising precipitation data from the preceding thirty years using a normal distribution centred on zero and its standard deviation; negative values represent dry periods.

WEFE nexus Water–Energy–Food–Ecosystem nexus, a framework describing the interdependencies between water, energy, food, and ecosystem health.